

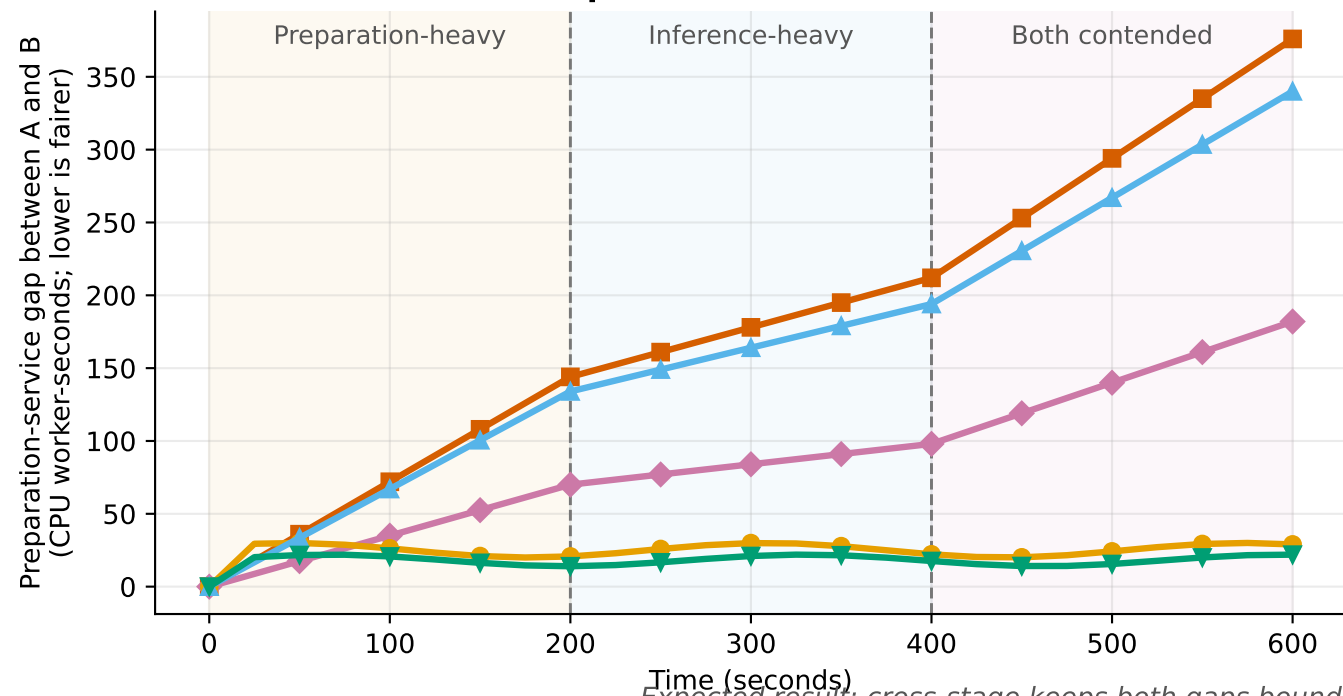
MOCKUP 8 — Fairness Under Shifting Multimodal Contention

Cross-stage scheduling keeps tenant service imbalance bounded at both stages

FCFS Tenant RR Prep-only Inference-only Cross-stage

Workload: two equal-rate, continuously backlogged tenants. Phase 1 varies frame cost; Phase 2 varies output cost; Phase 3 varies both. Identical trace across policies and seeds.

(a) Preparation-service fairness



(b) Inference-service fairness

